

University of Pisa

Department of Computer Science

Master's Degree in Data Science and Business Informatics

Optimization for Data Science

Convex Quadratic Separable Minimum-Cost Flow Problem

Project 37 Report

(P) is the convex quadratic separable Min-Cost Flow Problem

$$\min \left\{ x^T Q x + q x : Ex = b,\, 0 \le x \le u \right\}$$

where $E \in \mathbb{R}^{n \times m}$ is the node-arc incidence matrix of a directed connected graph with n nodes and m arcs, and $Q \in \mathbb{R}^{m \times m}$ is diagonal and Positive Semidefinite (i.e., $Q = \mathrm{diag}(Q) \ge 0$).

(A1) is a solution algorithm of the class of active-set methods.

(A2) is a general-purpose solver applied to an appropriate formulation of the problem.

No off-the-shelf solver is allowed, save of course for (A2).

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1 Introduction

This project focuses on the solution of a convex quadratic separable Minimum-Cost Flow problem defined over a directed connected network. This section delves into the properties of the problem, explains the overall structure of the equation and its constraints.

Let

$$G = (V, A)$$

be the directed connected graph, where V is the set of n nodes and A is the set of m directed arcs. A decision variable

$$x_j, \qquad j = 1, \ldots, m,$$

is associated with each arc and represents the amount of flow sent through that arc. Therefore,

$$x \in \mathbb{R}^m$$

denotes the complete vector of arc flows.

The optimization problem combines the classical structure of network flow with a convex nonlinear cost, subject to flow-conservation constraint and lower and upper bounds on the arc flows.

1.1 Objective

The main objective of the project is to study and solve the following convex quadratic separable Minimum-Cost Flow problem:

$$\begin{aligned} \min_{x \in \mathbb{R}^m} \quad & x^T Q x + q^T x \\ \text{s.t.} \quad & Ex = b, \\ & 0 \le x \le u, \end{aligned} \tag{1}$$

where $E \in \mathbb{R}^{n \times m}$ is the node–arc incidence matrix of the network, $b \in \mathbb{R}^n$ is the vector of node supplies and demands, $u \in \mathbb{R}^m$ contains the arc capacities, $q \in \mathbb{R}^m$ contains the coefficients of the linear transportation costs, and $Q \in \mathbb{R}^{m \times m}$ is a diagonal positive semidefinite matrix, and it is the coefficent of the quadratic terms. It can be written as:

$$Q = \operatorname{diag}(Q_1, \ldots, Q_m), \qquad Q_j \ge 0.$$

Two solution approaches are considered. The first approach, denoted by A1, is based on an active-set strategy specifically implemented for the problem. The second approach, denoted by A2, formulates the same optimization problem for a general-purpose optimization solver.

The project therefore has two closely related goals. First, it aims to implement a correct optimization algorithm capable of exploiting the constrained quadratic structure of problem (1). Second, it aims to experimentally investigate its computational behaviour and compare it with the general-purpose approach.

1.2 The Optimization Problem

Problem (1) can be interpreted as a nonlinear extension of the classical Minimum-Cost Flow problem and the objective function can be expressed as

$$f(x) = x^T Q x + q^T x = \sum_{j=1}^{m} \left(Q_j x_j^2 + q_j x_j \right).$$

Each arc therefore contributes independently to the total objective through the cost function

$$f_j(x_j) = Q_j x_j^2 + q_j x_j.$$

Although the cost is separable across the arcs, the decision variables are coupled by the network flow-conservation constraints. The optimizer must consequently determine how the available flow should be distributed over the network while simultaneously satisfying node balances, arc capacities, and minimizing the overall transportation cost.

1.2.1 Network Structure and Feasibility Constraints

The equality constraint

$$Ex = b$$

encodes the conservation of flow across the network.

For an arc $j = (r, s)$ directed from node r to node s, its corresponding column in E contains one nonzero entry associated with the tail node and one associated with the head node, with opposite signs.

A necessary consistency condition for the existence of a feasible flow is

$$\mathbf{1}^T b = 0,$$

meaning that the total amount supplied by the network must equal the total amount demanded.

For a connected graph, the node–arc incidence matrix satisfies

$$\operatorname{rank}(E) = n - 1,$$

since its rows are linearly dependent (for a detailed explanation, refer to Appendices A.1 and A.2). Given that active set optimization method is based on full rank matrices, one flow-conservation equation is redundant and may be removed.

In addition to flow conservation, each arc is subject to the capacity constraints

$$0 \le x_j \le u_j, \qquad j = 1, \dots, m.$$

The lower bound prevents negative flow on a directed arc, whereas u_j represents the maximum amount of flow that can be routed through arc j.

The feasible region is therefore

$$\mathcal{F} = \{x \in \mathbb{R}^m : Ex = b,\ 0 \le x \le u\}.$$

Since both the equality constraints and the bound constraints are linear, they define convex sets. Since the intersection of convex sets is convex, the feasible region $\mathcal{F}$ is convex.

1.2.2 Convexity and Separability of the Objective Function

The gradient of the objective function is

$$\nabla f(x) = 2Qx + q,$$

while its Hessian is

$$\nabla^2 f(x) = 2Q.$$

Since Q is positive semidefinite,

$$Q \succeq 0,$$

and therefore

$$\nabla^2 f(x) \succeq 0.$$

It follows that f is a convex function.

An additional important property is the separability of the objective. Since Q is diagonal, no cross-product terms of the form $x_i x_j$, $i \neq j$, appear in the objective. Thus, changing the flow on one arc directly modifies only the cost associated with that arc.

The variables nevertheless remain indirectly coupled through the flow-conservation equations $Ex = b$.

Since the feasible region is convex and the objective function is convex, problem (1) is a convex quadratic programming problem, consequently, a feasible point satisfying the appropriate optimality conditions is a global solution of the problem.

1.2.3 Optimality Conditions

Given that the problem is subject to multiple constraints, its Karush–Kuhn–Tucker (KKT) conditions need to be considered. These conditions characterize constrained optimality by requiring, among other properties, that no feasible descent direction exists, i.e., that the objective function cannot be locally improved without violating the constraints.

Let

$$\lambda \in \mathbb{R}^n$$

be the Lagrange multiplier associated with the flow-conservation constraints, while

$$\mu \geq 0 \qquad \text{and} \qquad \nu \geq 0$$

denote the multipliers associated with the lower and upper bounds, respectively.

Writing the bound constraints as

$$-x \leq 0, \qquad x - u \leq 0,$$

the Lagrangian is

$$\mathcal{L}(x, \lambda, \mu, \nu) = x^T Q x + q^T x + \lambda^T (Ex - b) - \mu^T x + \nu^T (x - u).$$

Stationarity with respect to x gives

$$2Qx + q + E^T \lambda - \mu + \nu = 0.$$

The complete KKT conditions can therefore be grouped into stationarity, primal feasibility, dual feasibility, and complementary slackness as follows:

Stationarity:	$2Qx + q + E^T \lambda - \mu + \nu = 0,$
Primal feasibility:	$Ex = b, \qquad 0 \leq x \leq u,$
Dual feasibility:	$\mu, \nu \geq 0,$
Complementary slackness:	$\mu_j x_j = 0, \qquad j = 1, \ldots, m,$
	$\nu_j (u_j - x_j) = 0, \qquad j = 1, \ldots, m.$

These conditions provide the fundamental optimality characterization used by the solution method.

1.2.4 Active and Free Arcs

The bound constraints naturally partition the arcs into three categories.

An arc can be active at its lower bound,

$$x_j = 0,$$

free,

$$0 < x_j < u_j,$$

or active at its upper bound,

$$x_j = u_j.$$

Let

$$L = \{j : x_j = 0\},$$

$$U = \{j : x_j = u_j\},$$

and

$$F = \{1, \ldots, m\} \setminus (L \cup U)$$

denote the lower-active, upper-active, and free sets, respectively.

For a free arc $j \in F$, complementary slackness implies

$$\mu_j = \nu_j = 0.$$

Hence, the stationarity condition reduces to

$$(2Qx + q + E^T\lambda)_j = 0.$$

If the optimal sets L, U, and F were known in advance, the original bound-constrained problem could be reduced to an equality-constrained quadratic problem involving only the free variables.

Indeed, fixing

$$x_L = 0 \qquad \text{and} \qquad x_U = u_U,$$

the flow-conservation constraints become

$$E_F x_F + E_U u_U = b,$$

or

$$E_F x_F = b - E_U u_U.$$

The corresponding KKT system for the free variables takes the form

$$\begin{bmatrix} 2Q_{FF} & E_F^T \\ E_F & 0 \end{bmatrix} \begin{bmatrix} x_F \\ \lambda \end{bmatrix} = \begin{bmatrix} -q_F \\ b - E_U u_U \end{bmatrix}.$$

The main difficulty is that the optimal active set is not known a priori. It is for this reason that it is required the use fo active-set algorithm able to identify the partition iteratively.

Starting from a feasible point and an initial estimate of the active set, the algorithm repeatedly solves an equality-constrained quadratic subproblem over the currently free variables. If the resulting direction encounters one of the bounds, the corresponding constraint becomes active. Conversely, the Lagrange multipliers can indicate that an active constraint should be released.

This mechanism provides the basis for algorithm A1.

(planned to be described in detail in the following chapter)

A Appendix

A.1 Consistency condition: $\mathbf{1}^T b = 0$

Let

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n.$$

Since E is the node–arc incidence matrix of the network, the sum of the entries of each column is equal to zero, and hence

$$\mathbf{1}^T E = 0.$$

The flow conservation constraints are expressed as

$$Ex = b.$$

Multiplying both sides by $\mathbf{1}^T$ gives

$$\mathbf{1}^T E x = \mathbf{1}^T b.$$

Since $\mathbf{1}^T E = 0$, it follows that

$$0 = \mathbf{1}^T b.$$

Therefore, a necessary consistency condition for the existence of a feasible flow is

$$\mathbf{1}^T b = 0.$$

Equivalently,

$$\mathbf{1}^T b = \sum_{i=1}^{n} b_i,$$

so that

$$\sum_{i=1}^{n} b_i = 0.$$

From a network-flow perspective, this condition states that the total amount supplied by the nodes must be equal to the total amount demanded by the nodes.

A.2 Rank of the incidence matrix

As shown above,

$$\mathbf{1}^T E = 0.$$

The rows of E are linearly dependent since

$$\mathrm{row}_1(E) + \mathrm{row}_2(E) + \cdots + \mathrm{row}_n(E) = 0.$$

Hence, one row can always be expressed as a linear combination of the others. For example,

$$\mathrm{row}_n(E) = -\sum_{i=1}^{n-1} \mathrm{row}_i(E).$$

It follows that

$$\mathrm{rank}(E) \leq n - 1.$$

If the graph is connected, there are no further linear dependencies among the rows of the incidence matrix:

$$\text{rank}(E) \geq n - 1.$$

Therefore,

$$\text{rank}(E) = n - 1.$$

Thus, although the flow-conservation system $Ex = b$ contains n equations, only $n - 1$ of them are linearly independent.